Excel Capstone Project-E-commerce Store Analysis

Project Objective:

Analyze, clean and visualize the given Data Set using Excel Formulas, Functions, Pivot Table, Charts and Dash Boards.

Dataset Description:

Sheet1 - Customer_Dim:

Customer_ID	Used FIND and REPLACE (c- to cust)
Name	Used TRIM and PROPER function
Country	Used FIND and REPLACE
Loyalty_Level	Used IFISBLANK for Missing values

Sheet 2 - Product_Dim :

Product_ID	Used FIND and REPLACE option
Cost	Found average and total cost using formula
Stock	Found total stock using formula
Cost & stock	Found missing values by IFISBLANK AND EXCEL FORMULAS

Sheet 3 – Store_Dim

Done HLOOKUP and VLOOKUP to merge table with **Sales_Fact.**

Highlighted merged columns in Green color in the sales_fact table

Sheet 4 - Sales_Fact

Order_Date	Used date format	
Quantity	USED IF,ISBLANK,MODE FOR MISSING QUANTITY	
Unit Price	USED IF,ISBLANK,AVERAGE FUNCTIONSFOR MISSING PRICE	
Discount	USED IF,ISBLANK,MODE FORMISSING DISCOUNT	
Total price	Used IF VLOOK TEXTformulas to calculate total price after discount	

	Used count function and excel formulas for calculations and counting.	

Data cleaning & transformation:

- Done cleaning ,filtering ,merging, sorting and made each column according to their formats eg.currency,text,number...
- Made all table into a single table using HLOOKUP,VLOOKUP,INDEX,TEXT for creating Pivot table.
- Differentiated each meriging table for identification in different colors.

Visualization and Insights

- **Regional Performance Imbalance:** The **Sales in each Region** line chart shows a significant drop-off in the "West" region. While East, North, and South maintain relatively high and stable sales, the West is underperforming.

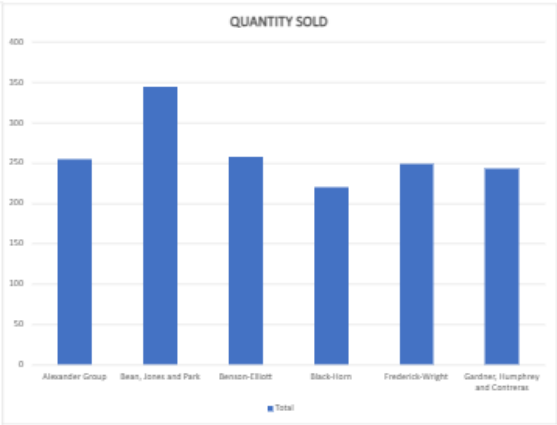
- **Top Product Performance:** The **Quantity Sold** bar chart indicates that products like "Amazon Echo Dot" and "Apple iPhone" are your volume drivers, significantly outperforming others in the list.

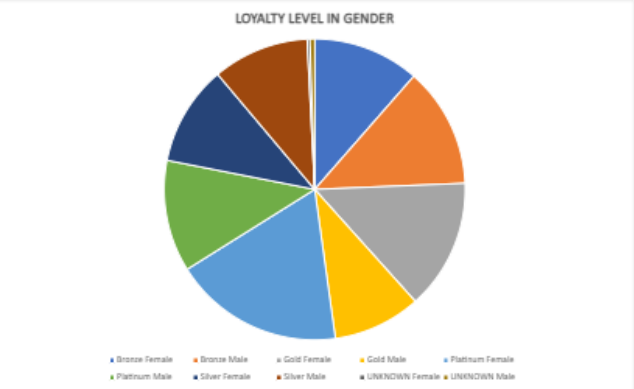
- **Gender-Based Loyalty:** The **Loyalty Level in Gender** pie chart suggests a diverse mix, but its complexity makes it hard to see which specific segment (e.g., "Gold Female" vs. "Gold Male") is the largest contributor.

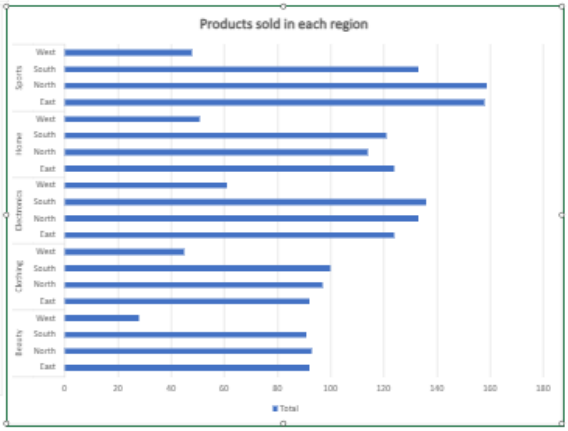
- **Regional Product Variety:** The **Products sold in each region** horizontal bar chart shows that the "East" region consistently has the highest variety and volume of products sold across multiple categories.

DASHBOARD

Sales in each Region
₹ 700,000.00
₹ 600,000.00
₹ 500,000.00
₹ 400,000.00
₹ 300,000.00
₹ 200,000.00
₹ 100,000.00
₹ -
East
North
South
West
Axis Title
Total


QUANTITY SOLD
400
350
300
250
200
150
100
50
0
Alexander Group
Bean, Jones and Park
Benson-Elliott
Black-Horn
Frederick-Wright
Gardner, Humphrey and Contreras
Total


LOYALTY LEVEL IN GENDER
Bronze Female
Bronze Male
Gold Female
Gold Male
Platinum Female
Platinum Male
Silver Female
Silver Male
UNKNOWN Female
UNKNOWN Male


Products sold in each region
Sports
Home
Electronics
Clothing
Beauty
West
South
North
East
0
20
40
60
80
100
120
140
160
180
Total